

Proposal: Comparative Topology of Multiple Onco-PPI Networks

Complex Networks

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1 Idea

The aim of the project is to analyze the human Protein-Protein interaction (PPI) networks to reveal how three major human cancers (breast, lung and ovarian) manifest at protein level. We aim to map these specific diseases onto the global protein network to test whether disease-associated proteins group into different functional communities (disease modules). Furthermore, we want to evaluate the structural vulnerability modules and model how biological dysfunction propagates through each specific cancer network.

2 Proposed Methodology

A comparative study will be conducted to answer the following research questions:

- What are the structural properties of each cancer-specific network? This question involves exploring a wide variety of fundamental network metrics, including average path length, clustering coefficient, etc. Moreover, see if they are all scale-free networks, or if there are different types of degree distribution depending on the cancer type.
- Which of the three networks exhibits a more defined community structure? The objective is to know whether the selected types of cancer are triggered by a group of proteins that interact among all of them, or instead there are highly distinguished and isolated communities.
- How do the networks respond to random failures and targeted attacks? A network may be robust to failures (i.e. eliminations of random nodes) but susceptible to directed deletions of specific nodes (selected using some kind of prior knowledge about the network, for instance the main hubs or biologically relevant proteins). The idea is to see how the topology of the network changes in each case and analyze how disease functions and protein interactions are altered. These types of conclusions can provide important insights. For instance, the identification of potential targets for drug discovery or understanding the underlying disease mechanisms.
- Is there a high overlap in the proteins involved in each type of cancer? We also aim to study if there exist proteins that act as *bridges* between the three networks, or if each cancer has their own specific molecules.
- Based on the current network topology, which PPI are likely but may not have been discovered yet? The idea is to analyze if the different cancer networks grow by preferential attachment and create centralized structures with hubs, or by common neighbors and creates cliques.

3 Data Source

The project will use data from the STRING database [1], given that it provides protein-protein interaction networks for *Homo sapiens*. First, we downloaded the global network and filtered it to keep only the highest-confidence interactions, using a cutoff threshold of 0.7. Thus, we only consider connections between proteins when STRING assigns a high enough confidence.

Subsequently, instead of analyzing the whole human network, we decided to focus on three cancer types: breast cancer, lung cancer and ovarian cancer. To do so, we used the STRING app [2] within Cytoscape [3] and imported each network as a disease query, entering the corresponding cancer name. We also set a maximum number of 2000 proteins for each cancer type.

Finally, we exported each network in .sif format, which has this simple structure:

```
protein1    pp    protein2
```

where `protein1` and `protein2` are proteins and `pp` indicates a protein-protein interaction. This format enables us to load the networks with Python and NetworkX to conduct the analysis.

4 Brief Data Analysis

An initial descriptive analysis of the networks associated with each cancer type has been conducted. Table 1 summarizes the main structural properties of each network.

Network	Nodes	Edges	Density	Avg. degree	Components	Largest CC	Isolates
Breast cancer	1812	33421	0.0204	36.89	34	1777	31
Lung cancer	1728	31398	0.0210	36.34	45	1684	44
Ovarian cancer	1761	32077	0.0207	36.43	38	1720	33

Table 1: Summary of the structural properties of the cancer-specific protein-protein interaction networks.

First, it is observed that the three networks present a similar size, around 1700 to 1800 nodes and more than 30000 interactions. The density of the networks is relatively low (around 0.02), which implies that we are dealing with sparse networks, as is commonly observed in real biological systems.

The average degree of our networks is high (around 36), suggesting that, despite the low density, there exist nodes with a considerable number of connections. These findings are consistent with the presence of hubs, that is, highly connected proteins.

Regarding connectivity, the networks are not completely connected, given that they present multiple connected components. However, in all cases there exists a giant connected component that groups the majority of the nodes (more than 95% in each network), indicating that most proteins are indirectly connected to each other. Moreover, isolated nodes have also been identified, although their number is small compared to the total size of the network.

Overall, these properties suggest that the protein-protein interaction networks associated with each cancer type present the typical structure of complex networks.

5 References

- [1] D. Szklarczyk, R. Kirsch, M. Koutrouli, K. Nastou, F. Mehryary, R. Hachilif, A. L. Gable, T. Fang, N. T. Doncheva, S. Pyysalo, P. Bork, L. J. Jensen, and C. von Mering. The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51(D1):D638–D646, Jan 2023.
- [2] Nadezhda T Doncheva, John H Morris, Jan Gorodkin, and Lars J Jensen. Cytoscape stringapp: network analysis and visualization of proteomics data. *Journal of proteome research*, 18(2):623–632, 2018.
- [3] Paul Shannon, Andrew Markiel, Owen Ozier, Nitin S Baliga, Jonathan T Wang, Daniel Ramage, Nada Amin, Benno Schwikowski, and Trey Ideker. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome research*, 13(11):2498–2504, 2003.